

Project - 1: Railway Guages and Data Analysis

This guide teaches you to CSV data, **break down** the facts, and **draw** charts to explain them visualize results using Matplotlib.

Pandas theory :

Pandas is a powerful data analysis library used for data manipulation and analysis. It provides DataFrame structures for handling tabular data. Pandas is widely used for data analysis because it provides simple and flexible methods to handle structured data.

First, we need to import the required libraries and load the dataset.

```
import pandas as pd
import matplotlib.pyplot as plt

# Using a raw string r'' helps avoid path errors we saw earlier
df = pd.read_csv("railway_gauges.csv")
```

Pandas converts the data into a **DataFrame**, which makes it easy to view and manipulate data. Using **head()** we can see the first 5 rows, and similarly **tail()** can be used to view last 5 rows.

Useful Pandas Operations:

Pandas provides different ways to access data:

- **df.columns**: This accesses the Index object containing all your column names
- **.str**: This is a "vectorized string accessor." It allows you to apply Python string methods to every element in that Index at once, rather than writing a loop.
- **.strip()**: This is the specific string method that trims spaces from the start and end of a string.

```
# Clean column names just in case there are hidden spaces
df.columns = df.columns.str.strip()
```

It "cleans" your data. Often, when importing CSV or Excel files.

Data Visualization:

To better understand the data, we can represent it graphically. Here, a bar chart is used to show how railway tracks were installed over the years.

```
# Increase figure size so the bars/labels have room to breathe
ax = df.plot(x='Year', kind='bar', figsize=(12, 6))

plt.xticks(rotation=45) # 45 is usually easier to read than 70
plt.xlabel('Year')
plt.ylabel('Track Count')
plt.title('Gauges: Number of railway tracks installed per year')

plt.tight_layout() # Prevents labels from getting cut off in the PNG
plt.savefig('rail_gauges.png')
plt.show()
```

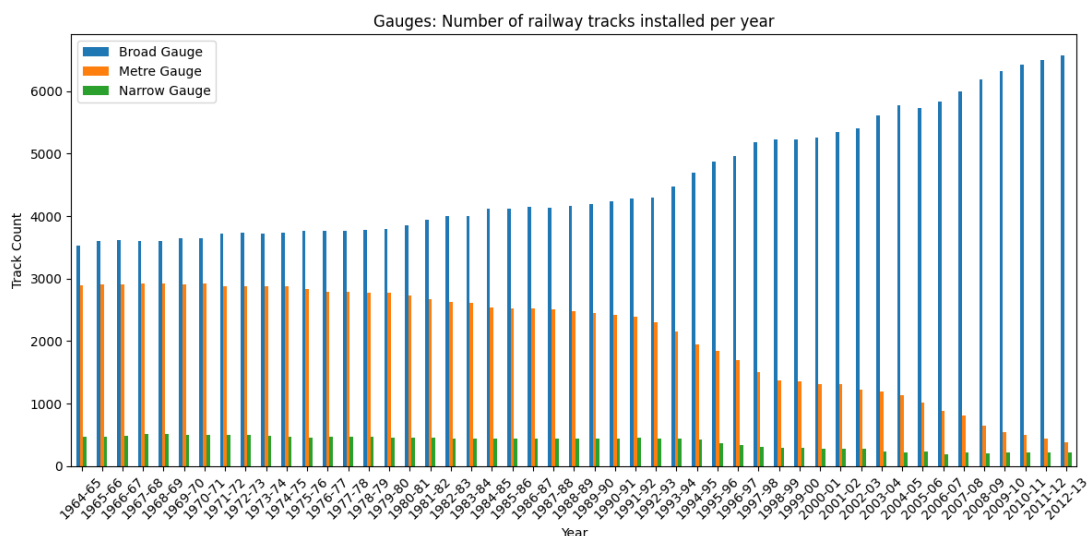
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The graph clearly shows the variation in the track installation across different years and gauge types.

Result:

- ✓ Identified the year with highest railway track installation.
- ✓ Visualized the data using a bar chart ✓ Understood how Pandas simplifies data analysis.

And the output looks like below.



Conclusion:

Through this project ,we learned how to:

1. Import and read CSV files
2. Perform basic data analysis
3. Extract useful insights
4. Represent data visually

This project helped me understand the data clearly and interpret the trends effectively using visualization.